Distinct Proteoform Profiles Mark Adult Zebrafish Telencephalon and Optic Tectum

Anonymous Authors

Abstract

The adult zebrafish telencephalon and optic tectum carry different neural work: forebrain integration, memory, and social behavior on one side; visual orientation and sensorimotor control on the other. We asked whether that division is visible in intact proteoforms in a published region-resolved LCM-CZE-MS/MS dataset. The shared repertoire was sparse. The article reported only 35 shared proteoforms among 807 total, and strict accession-plus-proteoform matching of the deposited Tel2 and Teo2 spreadsheets gave 29 shared exact IDs among 805. The split was not an artifact of notation, unequal regional list size, duplicate-level detectability, plausible identifier error, or protein and gene-symbol collapse. It also had biological direction: telencephalon favored neuropeptide and synaptic entries, whereas optic tectum favored reticulon, cytoskeletal, and motor-linked entries. A curated marker panel placed 125 of 128 observations in the expected region, and an independent literature-derived panel recovered the same axis. The modification layer was rich, but the apparent telencephalic excess of N-terminal acetylation weakened after covariate and first-residue checks. The durable conclusion is therefore simple: these two adult brain regions occupy distinct proteoform profiles that track known function, while broader protein summaries blur part of that regional order.

1 Introduction

Different brain regions do different work, and that work should leave traces in their molecular form. In the adult zebrafish, the telencephalon is tied to social behavior, learning, memory, and other forebrain functions (Stednitz et al., 2018; Reemst et al., 2023; Pandey et al., 2023). The optic tectum is organized around visual input, orienting, and sensorimotor control (Roeser and Baier, 2003; Orger, 2016). A protein-level summary can show part of this division. An intact-proteoform summary can show more, because sequence variants and post-translational modifications are not collapsed into a single protein name.

The LCM-CZE-MS/MS study of Lubeckyj and Sun (2022) gives a rare way to ask this question directly. It reported hundreds of top-down proteoforms from adult zebrafish telencephalon and optic tectum and linked several entries to regional function. The deposited PRIDE Tel2 and Teo2 spreadsheets make the comparison sharper. They allow each region to be read not only as a count of identifications, but as an accession-resolved set of proteoforms whose overlap, marker direction, and modification burden can be tested.

The answer is striking. Telencephalon and optic tectum share very few intact proteoforms. The entries that distinguish them fall along a familiar biological axis: neuropeptide and synaptic proteins on the telencephalic side, reticulon and cytoskeletal families on the tectal side. The stronger statement is not that every proteoform can be assigned a function, nor that every modification is regionally programmed. It is that two anatomically and functionally distinct regions also differ as intact-proteoform profiles.

This distinction matters for spatial proteomics. Imaging-based approaches preserve direct spatial context, whereas microdissection followed by separation and tandem mass spectrometry trades spatial pixels for deeper molecular coverage. In that bargain, identifier granularity becomes central. Exact accession-plus-proteoform identities, canonicalized proteoform strings, protein accessions, and gene symbols do not carry the same biological information. The present analysis shows how much regional order is lost as the identifier is relaxed.

Recent work has continued to show that zebrafish brain proteoforms contain age- and region-relevant information (Askarani et al., 2026). What has been missing is a compact, traceable account of regional sharing and separation in a public adult-brain top-down dataset. The evidence here supports a plain conclusion: adult zebrafish telencephalon and optic tectum are not simply neighboring samples of the same local proteome. They are marked by different proteoform profiles, and those profiles retain a recognizable imprint of what the regions do.

2 Materials and Analysis

2.1 Source material

We analyzed the adult zebrafish telencephalon and optic-tectum LCM-CZE-MS/MS dataset reported by Lubeckyj and Sun (2022). The material consists of the article, its supplementary workbook, and two region spreadsheets deposited in PRIDE, `Proteoform_ID_Tel2.xlsx` and `Proteoform_ID_Teo2.xlsx`. The supplement reports TopPIC identification at 1% proteoform-spectrum-match FDR and 5% proteoform-level FDR, followed by duplicate label-free quantification in TopDiff. The public files expose duplicate technical runs for each region; they do not expose a biological-replicate design across animals. The `Tel2` and `Teo2` sheets in the supplement match the PRIDE files exactly, so the region spreadsheets are the quantitative source used here, and the supplement supplies acquisition and search context.

The main comparison unit is the accession-plus-proteoform pair. This unit is deliberately strict. The same proteoform-like string can appear under more than one accession, and accession identity is therefore retained in the primary overlap analysis. Protein-level and gene-symbol summaries are treated as coarser sensitivity views. Marker families were specified from the biological language of the original report: exact gene-symbol matches for NPY, PENKB, PYY-A, and neurogranin a, and description-keyword matches for synucleins, reticulon, myosin, actin, and troponin. The local script `results/analyze_region_proteome.py` generates the mapping rules, full marker-membership tables, canonicalization examples, a raw-to-canonical map, cross-accession ambiguity diagnostics, and source-to-claim traceability.

2.2 Derived metrics

Three questions guided the analysis. First, how much of the intact-proteoform repertoire is shared between the two regions? Second, do region-enriched marker families follow known telencephalic and tectal biology? Third, does the apparent N-terminal acetylation difference survive simple checks for unequal recovery? Regional separation was assessed with article-level counts, exact accession-plus-proteoform overlap, canonicalized representation diagnostics, protein and gene-symbol collapse, abundance-weighted similarity, fixed-margin overlap tests, count-normalized rarefaction, duplicate-based detectability correction, and conservative misidentification bounds keyed to the published FDR values.

Biological direction was assessed twice. The first panel used the marker families emphasized in the original report. The second used an independent literature-derived composition panel, so that the result would not rest only on the original examples. Robustness checks included alternative abundance normalizations, duplicate-informed missingness bounds, minimal MNAR-style low-intensity fills, family-size-preserving randomization, conservative spillover reassignment, motor-family exclusion, protein collapse, gene-symbol collapse, rarefaction of the larger optic-tectum list down to the telencephalon count, and restriction to the top-intensity rows in each region. A targeted tissue-sentinel screen counted skeletal/cardiac muscle, contractile cytoskeleton, neuronal/synaptic, glial/myelin, and housekeeping panels by row count and mean duplicate intensity. The duplicate-based overlap correction was repeated after stratifying exact IDs by precursor-mass tertiles and by mean-intensity tertiles within each region.

Acetylation was handled as a boundary case rather than as a headline claim. The analysis was restricted to matched marker subsets and then examined with two coarse logistic sensitivity models: one with region as the outcome and acetylation as a predictor, and one with acetylation as the outcome and region as the predictor. Both models used log precursor mass, sequence span length, and first-residue position as covariates, followed by a restricted check on rows whose first residue lay at or within two amino acids of the N terminus. The formulas and implementation details are given in the Appendix.

2.3 Interpretive scope

The public files support overlap analysis, marker-family mapping, and conservative PTM sensitivity checks. They do not expose processed feature histories, localization-confidence records, or the full structure needed for a raw feature-level error model. The strongest claims are therefore about regional separation and the biological direction of that separation. On those questions, the evidence is direct.

3 Results

3.1 Telencephalon and optic tectum are sharply separated at the proteoform level

The first result is the simplest one. At intact-proteoform resolution, the two regions share little. The article reported 309 telencephalic proteoforms and 533 tectal proteoforms, with only 35 shared between them. Strict accession-plus-proteoform matching of the PRIDE Tel2 and Teo2 spreadsheets gives an even smaller shared set: 29 shared entries among 805 exact IDs. Coarser summaries increase the apparent overlap, as expected, but they do not remove the regional split. Gene-symbol collapse raises the Jaccard value to 0.2302, still far below its fixed-margin independence baseline (centered Jaccard -0.2034, $p = 6.29 \times 10^{-28}$). The two regions are therefore not small variants of a common local proteome.

The result is stable to representation. Across five abundance-normalization schemes, weighted Jaccard remains between 0.0427 and 0.0557, and Bray–Curtis similarity remains between 0.0820 and 0.1055. Fixed-margin testing places the observed exact-ID overlap well below the overlap expected from the two regional list sizes alone (expected Jaccard 0.3152, 95% null interval 0.2890–0.3430, centered Jaccard -0.2792, $p = 5.56 \times 10^{-183}$). The small difference between the article's 35 shared proteoforms and the spreadsheet-derived exact-ID count of 29 appears to be notational rather than biological. Relaxed matching that removes bracketed PTM annotations and punctuation while preserving accession identity raises the shared count to 44, but the overlap remains low (canonicalized Jaccard 0.0615, centered Jaccard -0.2627, $p = 5.01 \times 10^{-143}$).

The same conclusion survives the missingness checks. Duplicate-informed bounds, jackknife estimates, and minimal MNAR-style fills never make the pair a high-overlap case. A duplicate-rediscovery occupancy model raises exact-ID Jaccard only from 0.0360 to 0.0432 after correcting for uneven recovery. Stratifying that correction by precursor-mass tertiles or by mean-intensity tertiles changes the adjusted Jaccard only to 0.0451 and 0.0445. Subsampling the larger optic-tectum list down to the telencephalon count gives a median exact-ID Jaccard of 0.0289, with a 95% sampling interval of 0.0185–0.0377. Restricting each region to its 100 highest-intensity rows gives 10 shared exact IDs and an exact-ID Jaccard of 0.0526; the marker-axis alignment in that top-intensity subset is 1.0000. Conservative identifier-error bounds are similarly small: even if 5% of region-unique exact IDs are reassigned in the direction most favorable to overlap, the Jaccard ceiling rises only to 0.0540. Unequal recovery, unequal list size, low-intensity identifications, and plausible identifier error soften the edge of the split. They do not change its character.

Table 1: Core regional separation metrics. The first two rows preserve the article-level summary, and the remaining rows use the PRIDE region spreadsheets plus detectability and identifier-error bounds.

Metric	Value
Article shared proteoforms	35
Article Jaccard overlap	0.0434
Exact-ID shared proteoforms	29
Exact-ID Jaccard overlap	0.0360
Weighted Jaccard overlap	0.0427
Protein overlap fraction	0.2151
Count-normalized exact-ID Jaccard median	0.0289
Occupancy-adjusted shared exact IDs	43.2
Occupancy-adjusted Jaccard overlap	0.0432
5% misidentification ceiling	0.0540

3.2 Marker families follow known regional biology

Low overlap need not have biological meaning. Here it does. The curated marker panel is concentrated almost entirely in the region expected from known zebrafish neurobiology. Telencephalon-associated markers contribute 23 matched counts and only 2 spillover counts. Optic-tectum-associated markers contribute 102 matched counts and only 1 spillover count. Across the full panel, 125 of 128 observations fall in the expected region. Because optic tectum contributes the larger raw proteoform list, the alignment is not a trivial consequence of having more tectal identifications.

The support is strong by several views. The panel yields an odds ratio of 1173 with an approximate 95% confidence interval from 102.0 to 13,495.1 and a one-sided exact p value of 5.12×10^{-22}. A family-size-preserving randomization reaches the same conclusion without depending on asymptotic assumptions: in 200,000 randomizations that preserved the nine family sizes and the global 25-versus-103 telencephalon/optic-tectum label totals, none matched the observed 125 correctly placed counts. All nine marker groups point in the expected direction. Protein collapse, gene-symbol collapse, intensity weighting, count-normalized subsampling, and top-intensity restriction retain the same axis.

The motor-heavy families must be read carefully. A detailed accession breakdown shows that 86.3% of the myosin and troponin observations carry skeletal- or cardiac-muscle-like annotations. A targeted sentinel screen confirms the concern: skeletal/cardiac muscle sentinels appear as 2 telencephalon rows and 89 optic-tectum rows, contributing 0.042% of telencephalon mean duplicate intensity but 11.0% of optic-tectum intensity. They cannot be allowed to bear the biological interpretation by themselves. Their removal is therefore a useful stress test: after excluding myosin, actin, and troponin altogether, the matched-region fraction remains 0.9310.

The same pattern survives sharper perturbations. Leave-one-out removal of any single marker family keeps the matched-region fraction between 0.9625 and 0.9911. Conservative spillover reassignment still leaves the panel strongly aligned. The expanded sentinel screen also argues against a contamination-only explanation. The muscle signal is real and tectal-heavy, but neuronal/synaptic sentinels are concentrated by intensity in telencephalon (96.98% of panel intensity), glial/myelin sentinels also favor telencephalon by count (28 versus 7), and the non-motor panels remain regionally structured.

An independent panel curated from the broader zebrafish literature reaches the same conclusion. Adult zebrafish myelin markers and optic-tectum radial-glia markers yield 32 matched counts and only 3 spillover counts, for an alignment fraction of 0.9143 and a one-sided exact p value of 6.68×10^{-4}. The panel is smaller and more composition-oriented, but it matters because it widens the result beyond the original

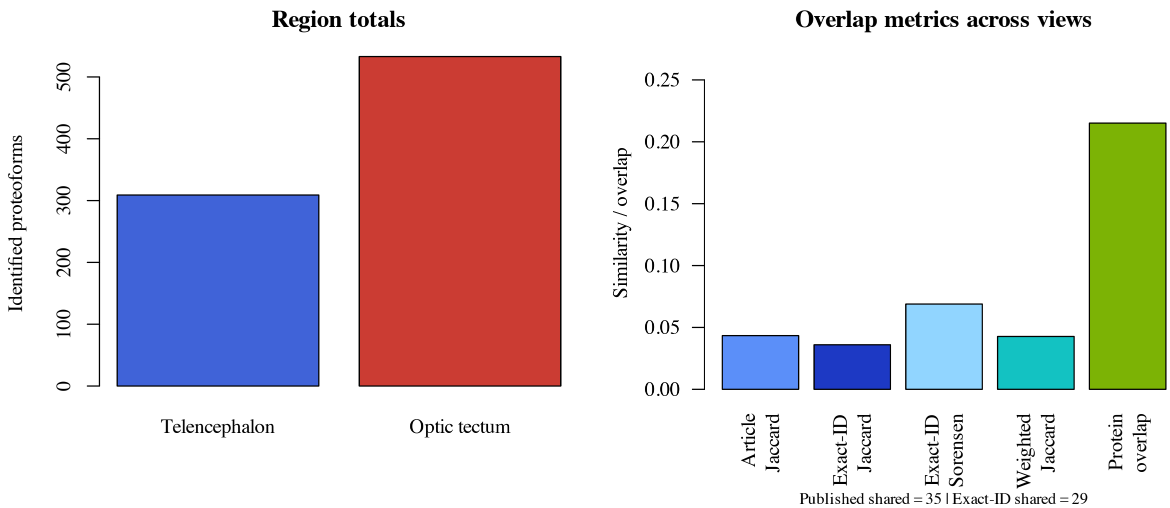

Figure 1: Regional totals and overlap metrics. The PRIDE region spreadsheets contain 303 exact IDs in telencephalon and 531 in optic tectum, with 29 exact-ID overlaps. This spreadsheet-based count is slightly stricter than the article aggregate, but both support the same low-overlap picture.

marker vocabulary. Count-normalized subsampling also preserves the original marker axis: when the larger optic-tectum row set is sampled down to the telencephalon row count, the median marker-alignment fraction is 0.9756.

3.3 The PTM layer is rich, but the acetylation contrast is not yet stable

The modification layer gives the paper a useful boundary. The original article reported 807 proteoforms, 358 with mass shifts and 211 with N-terminal acetylation. The material is plainly proteoform-rich, not merely a protein list in another format. But the specific acetylation contrast is weaker than the regional-overlap and marker-family results. In the matched marker subsets, 14 of 23 telencephalon-associated proteoforms are N-terminally acetylated, compared with 16 of 102 optic-tectum-associated proteoforms. After adjustment, that apparent difference fades. In a coarse logistic model with region as the outcome and acetylation, log precursor mass, sequence span length, and first-residue position as predictors, the adjusted odds ratio falls to 2.94 (95% CI 0.70–12.30; $p = 0.140$). The subset restricted to first residue ≤ 2 shows no enrichment. Reversing the model gives the same lesson: with acetylation as the outcome and region as the main predictor, the regional odds ratio is 0.26 with $p = 0.180$. Acetylation remains a lead, not a settled regional program.

The duplicate analyses explain why a small study can nevertheless support a clear regional result. Duplicate runs recovered a mean of 232 proteoforms in telencephalon and 356 in optic tectum, corresponding to recovery fractions of 0.7508 and 0.6680 relative to the regional totals. The design is limited, but the regional split is not a single-run accident.

4 Discussion

The adult zebrafish telencephalon and optic tectum carry different proteoform profiles. The evidence comes from several directions: sparse exact-ID overlap, coherent marker-family direction, abundance-aware similarities, duplicate-informed missingness checks, and conservative error bounds that never turn the two regions into a high-overlap pair.

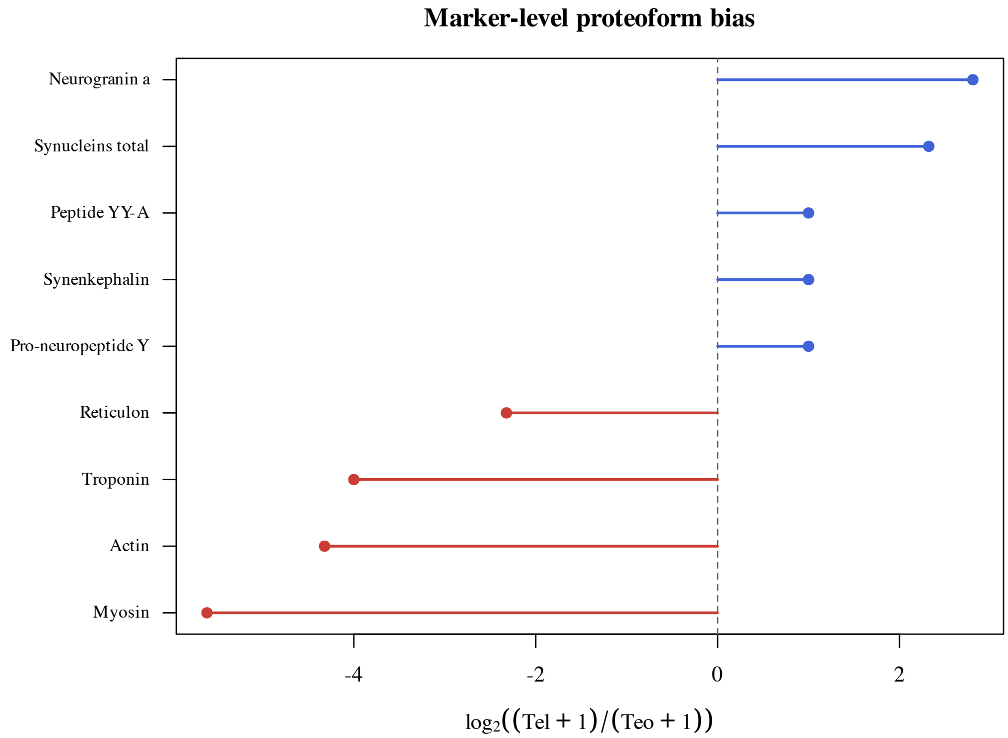

Figure 2: Marker-level proteoform bias across 128 curated proteoform observations in nine marker families. Positive values favor telencephalon and negative values favor optic tectum.

The magnitude of the split would matter less if it were biologically anonymous. It is not. Telencephalon is enriched for neuropeptide and synaptic entries consistent with a forebrain region involved in integration, learning, memory, and behavioral regulation. Optic tectum is enriched for reticulon, cytoskeletal, and motor-linked entries consistent with retinotectal organization and visually guided behavior. The two regions differ in a way that remains intelligible against established regional function.

The motor-associated families require restraint. Most myosin and troponin observations have skeletal- or cardiac-muscle-like annotations, and the expanded sentinel screen confirms a tectal-heavy muscle signal. They cannot carry a tectal interpretation on their own. The important observation is that the regional axis survives without them. After the motor-associated families are removed, the remaining marker panel remains strongly aligned, the independent literature-derived panel points in the same direction, and the high-intensity-only analysis still shows sparse overlap.

Unequal identification counts also fail to explain the result. When the larger optic-tectum list is repeatedly sampled down to the telencephalon count, exact-ID overlap remains sparse and marker alignment remains high. Restricting each region to its most intense 100 rows reaches the same conclusion. The sentinel guardrails sharpen rather than erase the interpretation: muscle-like carryover is plausible in optic tectum, but myelin and neuronal/synaptic panels are not collapsed into the same direction, and the motor-heavy families are dispensable for the main alignment. Direct purity measurements would still strengthen the anatomical interpretation, but the available checks do not support a contamination-driven explanation for the whole regional split.

The protein-collapsed view shows why proteoform resolution matters. Protein overlap is broader than proteoform overlap, as it should be, yet the regional split remains visible even after protein and gene-symbol

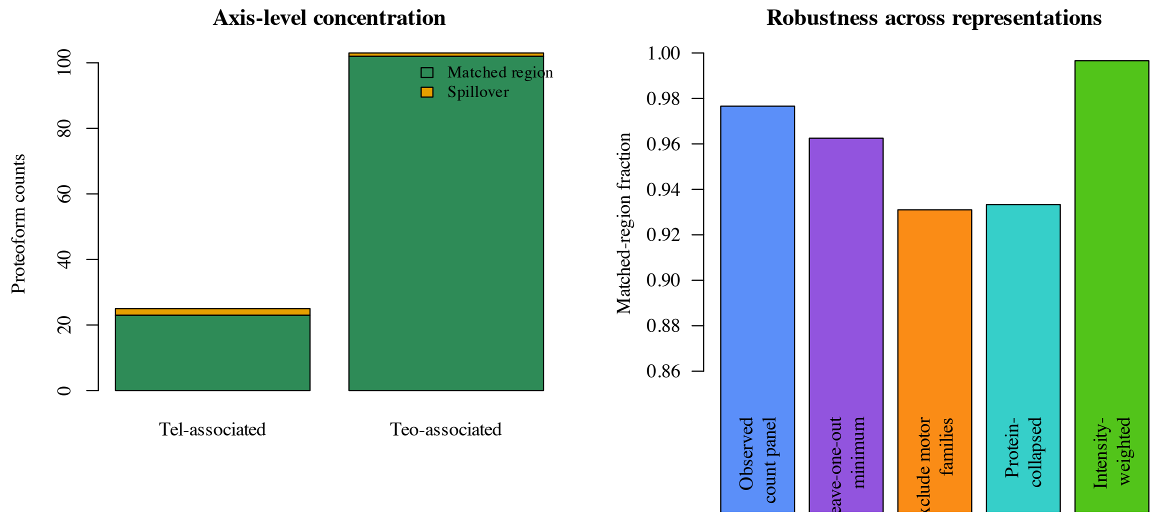

Figure 3: Axis-level concentration and robustness summaries for the nine curated marker families. The overall alignment signal remains high after leave-one-out perturbation, after excluding motor families, after collapsing to proteins, and after weighting by mean duplicate intensity.

collapse. Much of the biological distinction therefore lies in the resolved proteoform state rather than in simple protein presence or absence. For a region-resolved nervous-system dataset, that is the central lesson: the intact-proteoform layer preserves a regional texture that broader summaries partly dilute.

The result also clarifies the place of microdissected top-down proteomics. Imaging approaches such as LAESI and nano-DESI MSI preserve direct spatial context and on-tissue proteoform mapping (Kiss et al., 2014; Yang et al., 2022). LCM-CZE-MS/MS makes a different bargain. It sacrifices within-section imaging for deeper coverage within anatomically defined regions. That depth is what makes the overlap structure and marker architecture legible here.

The region spreadsheets carry a practical lesson about representation. The difference between the article's 35 shared proteoforms and the strict exact-ID count of 29 appears to arise from notation rather than biology. Relaxed matching changes the count, but not the inference. The full raw-to-canonical map contains 842 rows and 17 canonicalized sequence groups that appear under more than one accession, which is why the primary analysis retains accession identity. Standards such as ProForma help separate notational variation from biological difference (LeDuc et al., 2018, 2022). These files do not contain enough localization metadata for full ProForma 2.0 serialization of every entry, but they contain enough structure to show that the regional conclusion is stable across reasonable canonicalization choices.

More broadly, exact accession-resolved entries, canonicalized strings, protein accessions, and gene symbols do not tell the same story. Coarser identifiers broaden overlap step by step, but they do not eliminate the regional separation. This is why proteoform-family thinking matters in top-down proteomics: the biological meaning changes as the identifier is relaxed (Schaffer et al., 2018). It also explains why missingness and provenance matter. Current proteomics discussions have emphasized that data-analysis choices shape the conclusions readers can trust (Vanderaa and Gatto, 2023); the duplicate and rarefaction checks here make that dependence explicit.

The acetylation result is different. It weakens as soon as detectability-related covariates are modeled. That boundary is useful, because it marks the point at which these data stop warranting a strong biological claim. The durable finding is regional proteoform separation, not a settled regional PTM program.

Table 2: Curated marker families used in the axis-alignment analysis. Exact accession+proteoform members for each family are listed in the local marker-membership file, but the biological interpretation still follows the marker groups named in the original paper.

Marker group	Tel	Teo	Interpretation
Pro-neuropeptide Y	1	0	feeding-related neuropeptide signaling
Synenkephalin	1	0	affective signaling
Peptide YY-A	1	0	peptide hormone signaling
Neurogranin a	6	0	memory-linked synaptic signaling
Synucleins total	14	2	synaptic and mnemonic family
Reticulon	0	4	retinotectal organization
Myosin	0	48	motor machinery
Actin	0	19	cytoskeletal movement
Troponin	1	31	contraction-linked regulation

Table 3: Axis-level alignment compared with the naive baseline implied by regional prevalence.

Functional axis	Observed	Wilson 95% CI	Baseline	Lift
Telencephalon-associated	0.9200	0.7503–0.9778	0.1875	0.7325
Optic-tectum-associated	0.9903	0.9470–0.9983	0.8125	0.1778

4.1 Limits of inference

The claim is shaped by the available material: processed public tables rather than raw feature histories. The evidence is strongest where the files are most direct, namely regional overlap and the biological direction of marker families. The duplicate-labeled spreadsheets support missingness bounds, normalization checks, marker traceability, occupancy-style detectability correction, top-intensity restriction, tissue-sentinel screening, and a limited PTM screen. They do not expose the feature histories, localization confidence, or biological replication across animals that a raw TopPIC export and a larger experimental design would provide (Choi and Liu, 2022). Those limits narrow the claim. They do not weaken the central result.

The strongest extension would combine public raw files with processed search intermediates so that the same question could be asked at feature level and under richer missingness models (Karpievitch et al., 2012; Lazar et al., 2016). Additional brain regions would test how general this axis of separation is, and explicit purity measurements would strengthen the anatomical interpretation. The present data already secure the main point: adult zebrafish telencephalon and optic tectum are distinguished by different proteoform profiles, and those profiles track known regional biology.

5 Conclusion

Adult zebrafish telencephalon and optic tectum are separated by distinct proteoform profiles, not merely by small shifts in abundance. The separating entries follow the expected biology of forebrain integration on one side and visuomotor architecture on the other, whereas the apparent acetylation asymmetry remains provisional. The proteoform layer therefore preserves a biological order that protein-level summaries partly obscure. In these data, two neighboring parts of the adult brain carry different molecular signatures of the work they do.

Table 4: Sensitivity checks requested by the external review. The motor-family exclusion uses a Haldane correction because one spillover cell is zero after exclusion.

Scenario	Alignment	Odds ratio	One-sided p
Observed panel	0.9766	1173.0	5.12×10^{-22}
Shift 1 matched count per axis	0.9609	370.3	2.01×10^{-19}
Shift 2 matched counts per axis	0.9453	175.0	3.73×10^{-17}
Shift 3 matched counts per axis	0.9297	99.0	3.93×10^{-15}
Exclude myosin, actin, troponin	0.9310	84.6	6.32×10^{-4}

Table 5: Technical sensitivity values reported in the original study and extended with exact-ID duplicate-incidence counts from the PRIDE region spreadsheets.

Region	Mean	SD	CV	Recovery	exact-ID q_1	Chao2
Telencephalon	232	28	0.1207	0.7508	152	385.5
Optic tectum	356	88	0.2472	0.6680	351	887.0

References

Mehrdad Falamarzi Askarani, William Poulos, Maryam Rahimzadeh Dashtaki, Seyed Amirhossein Sadeghi, Jose B. Cibelli, Fei Fang, and Liangliang Sun. Top-down proteomics study of aging and sexual dimorphism of zebrafish brains. *Journal of Mass Spectrometry*, 61(1):e70009, 2026. doi: 10.1002/jms.70009.

In Kwon Choi and Xiaowen Liu. Top-down mass spectrometry data analysis using TopPIC suite. In *Top-Down Proteomics*, volume 2500 of *Methods in Molecular Biology*, pages 83–103. Humana, 2022. doi: 10.1007/978-1-0716-2325-1_8.

Yuliya V. Karpievitch, Alan R. Dabney, and Richard D. Smith. Normalization and missing value imputation for label-free LC-MS analysis. *BMC Bioinformatics*, 13(Suppl 16):S5, 2012. doi: 10.1186/1471-2105-13-S16-S5.

András Kiss, Donald F. Smith, Brent R. Reschke, Matthew J. Powell, and Ron M. A. Heeren. Top-down mass spectrometry imaging of intact proteins by laser ablation ESI FT-ICR MS. *Proteomics*, 14(10):1283–1289, 2014. doi: 10.1002/pmic.201300306.

Celine Lazar, Laurent Gatto, Myriam Ferro, Cécile Bruley, and Thomas Burger. Accounting for the multiple natures of missing values in label-free quantitative proteomics data sets to compare imputation strategies. *Journal of Proteome Research*, 15(4):1116–1125, 2016. doi: 10.1021/acs.jproteome.5b00981.

Richard D. LeDuc, Veit Schwämmle, Michael R. Shortreed, Anthony J. Cesnik, Stefan K. Solntsev, et al. ProForma: A standard proteoform notation. *Journal of Proteome Research*, 17(3):1321–1325, 2018. doi: 10.1021/acs.jproteome.7b00851.

Richard D. LeDuc, Eric W. Deutsch, Pierre-Alain Binz, Ryan T. Fellers, Anthony J. Cesnik, Jason A. Klein, Tim Van Den Bossche, Ruben Gabriels, Anoop Yalavarthi, Yasset Perez-Riverol, Jeremy Carver, Wout Bittremieux, Shin Kawano, Bryan Pullman, Nuno Bandeira, Neil L. Kelleher, Philip M. Thomas, and Juan A. Vizcaíno. Proteomics standards initiative's ProForma 2.0: Unifying the encoding of proteoforms and peptidoforms. *Journal of Proteome Research*, 21(4):1189–1195, 2022. doi: 10.1021/acs.jproteome.1c00771.

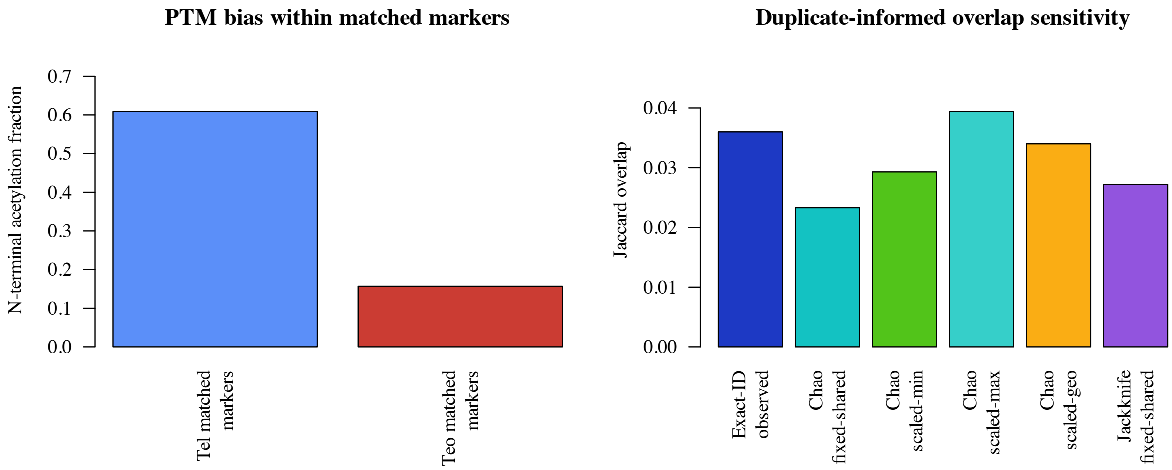

Figure 4: PTM and missingness-sensitive follow-up analyses. Left: N-terminal acetylation in the matched-marker subsets (23 telencephalon counts and 102 optic-tectum counts). Right: duplicate-informed Chao2 sensitivity changes the exact-ID Jaccard modestly, but never enough to erase the regional separation.

Rachele A. Lubeckyj and Liangliang Sun. Laser capture microdissection-capillary zone electrophoresis-tandem mass spectrometry (LCM-CZE-MS/MS) for spatially resolved top-down proteomics: a pilot study of zebrafish brain. *Molecular Omics*, 18(2):112–122, 2022. doi: 10.1039/d1mo00335f.

Michael B. Orger. The cellular organization of zebrafish visuomotor circuits. *Current Biology*, 26(9): R377–R385, 2016. doi: 10.1016/j.cub.2016.03.054.

Shristi Pandey, Anna J. Moyer, and Summer B. Thyme. A single-cell transcriptome atlas of the maturing zebrafish telencephalon. *Genome Research*, 33(4):658–671, 2023. doi: 10.1101/gr.277278.122.

Kitty Reemst, Heba Shahin, and Or David Shahar. Learning and memory formation in zebrafish: Protein dynamics and molecular tools. *Frontiers in Cell and Developmental Biology*, 11:1120984, 2023. doi: 10.3389/fcell.2023.1120984.

Tobias Roeser and Herwig Baier. Visuomotor behaviors in larval zebrafish after GFP-guided laser ablation of the optic tectum. *Journal of Neuroscience*, 23(9):3726–3734, 2003. doi: 10.1523/JNEUROSCI.23-09-03726.2003.

Leah V. Schaffer, Michael R. Shortreed, Anthony J. Cesnik, Brian L. Frey, Stefan K. Solntsev, Mark Scalf, and Lloyd M. Smith. Expanding proteoform identifications in top-down proteomic analyses by constructing proteoform families. *Analytical Chemistry*, 90(2):1325–1333, 2018. doi: 10.1021/acs.analchem.7b04221.

Sarah J. Stednitz, Erin M. McDermott, Denver Ncube, Alexandra Tallafuss, Judith S. Eisen, and Philip Washbourne. Forebrain control of behaviorally-driven social orienting in zebrafish. *Current Biology*, 28 (15):2445–2451.e3, 2018. doi: 10.1016/j.cub.2018.06.016.

Christophe Vanderaa and Laurent Gatto. The current state of single-cell proteomics data analysis. *Current Protocols*, 3(1):e658, 2023. doi: 10.1002/cpz1.658.

Manxi Yang, Hang Hu, Pei Su, Paul M. Thomas, Jeannie M. Camarillo, Joseph B. Greer, Bryan P. Early, Ryan T. Fellers, Neil L. Kelleher, and Julia Laskin. Proteoform-selective imaging of tissues using mass spectrometry. *Angewandte Chemie International Edition*, 61(29):e202200721, 2022. doi: 10.1002/anie.202200721.

A Technical Notes

A.1 Overlap and representation

The primary overlap unit is accession plus proteoform string. This preserves cases in which similar proteoform-like strings are assigned to different accessions. Protein-level and gene-symbol summaries are reported as coarser views, not as replacements for accession-resolved analysis.

Canonicalized matching was used only to diagnose representation effects. In that view, bracketed PTM annotations were removed with the pattern `\[[^\]]+\]`, parentheses and periods were stripped, and the remaining string was restricted to capital sequence letters `A-Z`. Accession identity was retained unless the analysis was explicitly protein- or gene-collapsed. The canonicalized comparison explains why the article-level count of 35 shared proteoforms lies between the strict exact-ID count and the relaxed representation, but it does not change the regional inference. The complete raw-to-canonical table is `results/canonicalization_full_map.csv`; accession conflicts are listed in `results/cross_accession_ambiguities.csv`.

Prevalence-adjusted overlap was evaluated with fixed telencephalon and optic-tectum margins. Let n_T and n_O be the two regional list sizes, N the union size, and X the observed shared count. The observed Jaccard is $J = X/(n_T + n_O - X)$. The marginal-prevalence baseline is

$$J_0 = \frac{p_T p_O}{p_T + p_O - p_T p_O},$$

where $p_T = n_T/N$ and $p_O = n_O/N$. The centered Jaccard is $J - J_0$. For uncertainty under fixed margins, the shared count was treated as $X \sim \text{Hypergeometric}(N, n_T, n_O)$, and the hypergeometric support was transformed to Jaccard values $x/(n_T + n_O - x)$. Under strict exact-ID matching, the observed Jaccard was 0.0360 against an independence baseline of 0.3152 and a 95% fixed-margin null interval of 0.2890–0.3430. Under canonicalized matching, the observed Jaccard was 0.0615 against a baseline of 0.3243. After protein collapse, the overlap rose to 0.2151 but still remained below the protein-level baseline of 0.4280.

A.2 Detectability and missingness

Duplicate `Match`/`No match` flags in the PRIDE region spreadsheets supplied the incidence counts used for Chao2 and first-order jackknife richness calculations. Blank intensity cells were treated as absent detections in the baseline analysis. A minimal MNAR-style sensitivity filled missing exact IDs to half of the region-specific 5th- or 10th-percentile nonzero intensity.

The duplicate-based detectability correction treated each exact ID as detected in one run or both runs within a region. The resulting two-run model estimates a region-specific per-run detection probability and an "observed at least once" probability. Shared exact IDs were then adjusted by the product of the two region-level detection probabilities. The same calculation was repeated after stratifying exact IDs by precursor-mass tertiles and by mean-intensity tertiles.

Misidentification sensitivity asked how much the exact-ID Jaccard could rise if an equal fraction of region-unique exact IDs in both regions were paired representation or identification errors. The 1% and 5% scenarios follow the PrSM- and proteoform-level FDR values reported in the supplement. Even the 5% scenario left the overlap low.

Identification-count rarefaction asked whether the larger optic-tectum list alone created the separation. In each of 10,000 trials, the larger regional set was sampled without replacement down to the smaller set size, with the seed fixed at 20260515. The calculation was repeated for exact accession-plus-proteoform IDs, canonicalized accession-plus-sequence IDs, protein accessions, and row-level marker alignment. The output is written to `results/identification_count_rarefaction.csv`.

Top-intensity restriction asked whether the low-overlap result depended on weak identifications. Each region was sorted by mean duplicate intensity, and the top 50, 100, and 150 rows per region were compared separately. The top-100 exact-ID Jaccard was 0.0526, with 10 shared exact IDs, and the marker-axis alignment fraction was 1.0000. The output is written to `results/top_intensity_restriction.csv`.

A.3 Marker families and acetylation

Marker alignment was evaluated with matched-versus-spillover counts, Wilson intervals, an approximate odds-ratio interval, an exact directional test, and family-size-preserving randomization. The randomization preserved the nine family sizes and the global telencephalon/optic-tectum label totals while shuffling labels across panel counts. Leave-one-family-out analysis, motor-family exclusion, protein collapse, gene-symbol collapse, count-normalized subsampling, and intensity weighting all retained the same regional direction.

A tissue-sentinel screen used review-requested skeletal/cardiac muscle terms, broader contractile-cytoskeleton terms, neuronal/synaptic terms, glial/myelin terms, and housekeeping terms. It reports row counts and mean duplicate intensity shares in `results/tissue_purity_sentinels.csv`. Row-level memberships are saved in `results/tissue_purity_sentinel_membership.csv`. This screen is a guardrail, not a substitute for direct histological purity measurement.

The acetylation analysis used matched marker rows only. One logistic model coded region as the outcome and used acetylation status, log precursor mass, sequence span length, and first-residue position as predictors. The inverse model coded acetylation as the outcome and region as the main predictor with the same covariates. A second analysis restricted rows to first residue ≤ 2 as a stricter N-terminal screen. These checks weakened the apparent telencephalic acetylation enrichment.

A.4 Limits of the public files

The journal supplement records that the region spreadsheets were generated with 1% PrSM FDR and 5% proteoform-level FDR in TopPIC and quantified across duplicate runs in TopDiff. The PRIDE landing page lists the four raw duplicate files, but the processed feature histories needed for a full raw-data rerun are not present. The analysis is therefore strongest for regional overlap, marker-family direction, and conservative PTM sensitivity. It is not a complete proteoform atlas of the adult zebrafish brain.

All derived tables in this manuscript are produced by `results/analyze_region_proteome.py`. The compact reproduction notebook is `code/reproduce_tables_and_figures.ipynb`. Canonicalization examples, full raw-to-canonical mappings, marker-family memberships, overlap tests, tissue sentinels, top-intensity restrictions, and the source-to-claim traceability record are saved under `results/` and `data/evidence_traceability.json`.